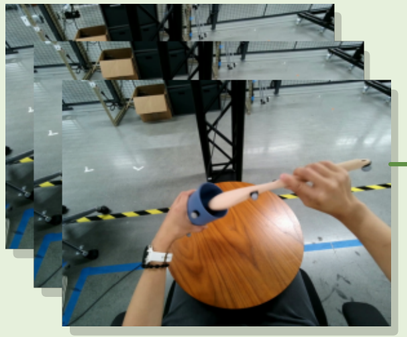
Egocentric Episode

raw video



colour camera + mono stereo pair

Task & Objects via VLM

Qwen3-VL (local)

"The person picks up a blue cup and a wooden spoon, stirs the contents of the cup with the spoon, and then places both items back on the table."

"wooden spoon", "blue cup"

Hand Estimator

Dyn-HaMR → **metric MANO trajectory** → **hands out of objects** → $\mathcal{M}_t^h$

WiLoR stereo fallback (<2%)

Metric Geometry

FoundationStereo → D_t

DROID-SLAM → T_t

table plane → gravity g

$D_t,\ T_t$

Object Estimator

SAM 3 (text-prompted) → $\mathcal{S}_t^o$ → **FoundationPose all ~252 hyp.** → $\{X_t^k\}$

SAM 3D Objects + stereo scale → $\mathcal{M}^o$ → **TrackEverything 3D tracks** → $p_n(t)$

SelectTrack: interaction-aware refinement

Whole-sequence selection

$$\min_{k_{1:T}} \sum_t u_t(\mathcal{S}_t^o, X_t^{k_t}, p_n) + \lambda\, m_t(p_n)$$

Speed-limit flip filter

State-aware guards

$s_t \in \{\text{resting, held}\}$

Contact sanity

4D HOI Reconstruction

$\{\mathcal{M}_t^h,\ X_t^o,\ \mathcal{M}^o\}$

Robot Learning

Simulator grounding

Retarget

SAC

Robot policy π